Daniel A. Delgado

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WORK EXPERIENCE

University of Florida **Gainesville, FL**

Enhancing Collaborator Communication in Extended Reality | XR Researcher *August 2024 - July 2025*

- Designed and evaluated shared-gaze visualization systems to support collaborative industrial-style tasks in augmented reality (AR) environments.
- Led end-to-end experimental pipelines (data collection, preprocessing, statistical analysis, visualization) across three controlled user studies (n > 60).
- Developed real-time AR prototypes integrating gaze tracking, spatial UI feedback, and performance metrics to improve task efficiency and coordination.
- Translated behavioral and performance data into actionable system design insights to optimize human–machine interaction.

ENKIx – Context-Aware Task Guidance | User Interface Designer *June 2021 - July 2024*

- Developed a context-aware AR task guidance system using Unity and C#/.NET to support step-by-step execution in constrained environments.
- Implemented computer vision–based object detection using ArUco markers to enable real-time spatial registration and system feedback loops.
- Collaborated with cross-functional engineering teams to integrate networking and biometric components into production-ready prototypes.
- Iteratively refined system performance and usability using quantitative metrics and user feedback.

EMG-Based Prosthetic Simulation in Virtual Reality | Full-Stack Developer *June 2020 - June 2023*

- Built an end-to-end ML pipeline for real-time EMG signal processing, feature extraction, model training, and inference.
- Implemented and evaluated linear regression and deep neural network models to classify hand and arm gestures from streaming sensor data.
- Optimized real-time inference performance to support closed-loop interaction in VR-based training simulations.
- Designed experimental protocols and performed data-driven analysis to compare system performance and user workload across virtual and physical platforms.

Characterizing User Comprehension in the STIR/SHAKEN Protocol | UX Researcher *June 2020 - June 2021*

- Conducted mixed-methods research to evaluate user interpretation of authentication states in telecommunications security systems.
- Designed surveys and interviews, analyzed quantitative and qualitative data, and translated findings into system-level recommendations.

Multi-Modal Affect Detection | Machine Learning Engineer *August 2019 - August 2020*

- Developed deep neural network architectures for multimodal emotion recognition using audio, image, and video data.
- Designed feature extraction pipelines (including wavelet transforms) and trained supervised ML models on large, heterogeneous datasets.
- Evaluated model performance across modalities and optimized fusion strategies to improve predictive accuracy.

EDUCATION

University of Florida **August, 2025**

PhD, Computer Science GPA: 3.6 / 4.0

- McKnight Doctoral Fellow
- Presented cutting-edge research at international conferences.

SKILLS

Machine Learning & AI: Deep learning, supervised learning, multimodal modeling, real-time inference, ML pipelines
Programming: Python, C#, C/C++, TensorFlow, Unity
Computer Vision: Image processing, object detection, sensor-based perception
Data & Analysis: Statistical analysis (R, MATLAB, Excel), experiment design, data visualization
UX & Systems: Human-centered design, usability testing, AR/VR prototyping, Physical AI interaction